Lucas Z. Brito

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Education

2020–2024.5 Brown University, Providence, RI. B.Sc. in Mathematical Physics and Computer Science

2014–2020 The Overlake School, Redmond, WA

Publications

2023 **L. Z. Brito**, S. Carr, J. A. Jacoby, J. B. Marston. *Hamiltonian Reconstruction: the Correlation Matrix and Incomplete Operator Bases,* Physical Review B. [PhysRevB.110.125118].

2022 D. Candoli, I. K. Nikolov, **L. Z. Brito**, S. Carr, S. Sanna, V. F. Mitrović, *PULSEE: A software for the quantum simulation of an extensive set of magnetic resonance observables*". [doi:j.cpc.2022.105898].

Employment

2023 *Instructional Lab Assistant, Brown Department of Physics*

Specialization in quantum optics. Designed and implemented tabletop demonstration of Bell's inequality, single photon interferometry. Wrote Python photon acquisition software for laboratory usage.

2022–2023 *Teaching Assistant, APMA1930W - Probabilities in Quantum Mechanics*

Wrote and lectured supplementary curriculum on quantum information theory and related topics. Substitute lectured to 30+ students.

2017–2018 *Assistant instructor, Play-Well Teknologies*

Skills

Programming Languages: Python (SciPy, NumPy, Tensorflow/PyTorch, Matplotlib, Pandas), Julia, Java, Javascript (React, Vue), HTML, CSS, SQL, LaTeX, bash scripting (Slurm, Git).

Languages: English, Portuguese, Spanish (conversational).

Awards and Recognition

2023–2024 Goldwater Scholar in Physics and Astronomy.

Relevant Coursework

Physics
- Advanced Electromagnetic Theory
- Quantum Mechanics I*, II*
- Advanced Quantum Mechanics*
- Advanced Statistical Mechanics*
- Computational Physics*
- Quantum Theory of Fields I*, II*
- Quantum Many-Body Theory*
- Gauge Theory*
- Standard Model*
- Biophysics*

Computer Science
- Accelerated Intro to Computer Science
- Data Science
- Intro to Software Engineering
- IS in Biophysical Scientific Visualization
- Interdisciplinary Scientific Visualization*
- Deep Learning*

Mathematics
- Linear algebra
- Partial Differential Equations
- Abstract Algebra
- Complex Analysis
- Differential Geometry*

(* = graduate level)

Outreach

2021–pres. Undegraduate Physics Journal Club, Co-Head.